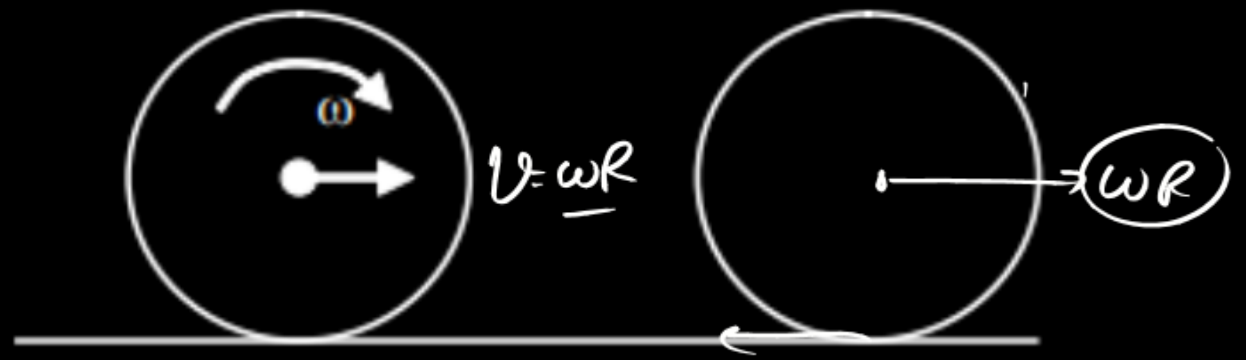
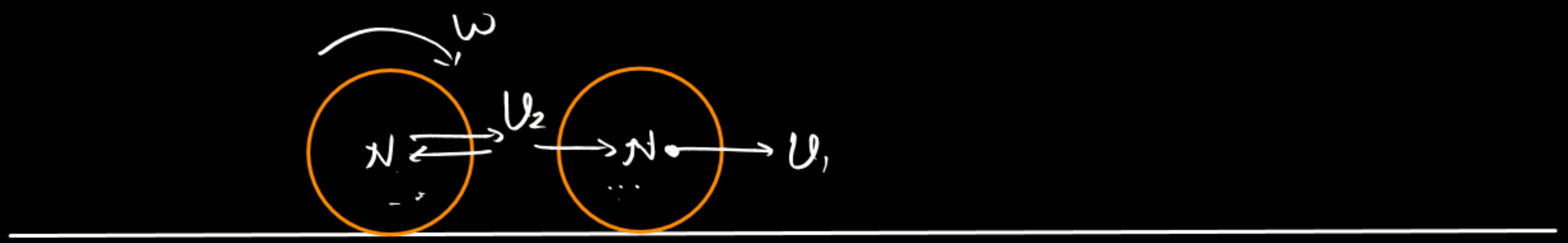
A solid sphere is rolling without slipping on rough ground as shown in fig. It collides elastically with an identical another sphere at rest. There is nsso friction between the two spheres. Radius of each sphere is 'R' and mass is 'M'.

no



Linear velocity of first sphere after it again starts rolling without slipping is

(A) $\frac{2}{5}\omega R$ (B) $\frac{2}{7}\omega R$ (C) $\frac{7}{10}\omega R$ (D) $\frac{7}{5}\omega R$



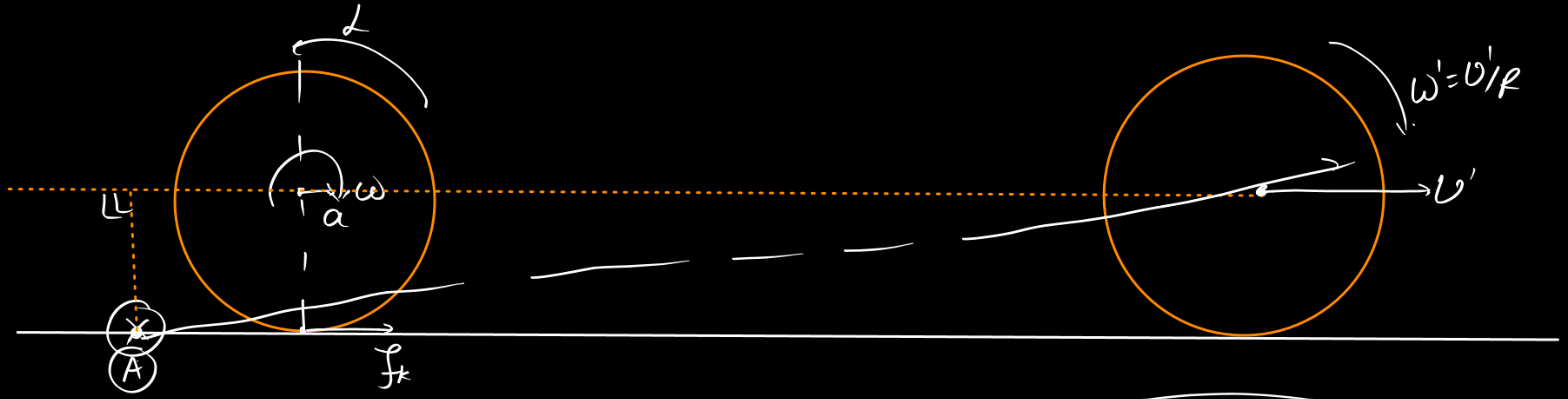
$$\cancel{m}(\omega R) + 0 = \cancel{m} v_2 + \cancel{m} v_1 \quad -(1)$$

$$e = \frac{v_1 - v_2}{\omega R} = 1$$

$$v_1 - v_2 = \omega R$$

$$v_2 = 0$$

$$v_1 = \omega R$$



$$L_i = 0 + \frac{2}{5} m R^2 \omega \otimes$$

$$L_f = m v' R \otimes + \frac{2}{5} m R^2 \omega' \otimes$$

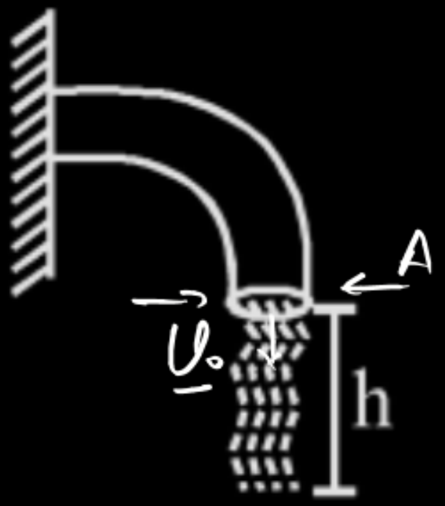
$$= m v' R + \frac{2}{5} m v' R = \frac{7}{5} m v' R \otimes$$

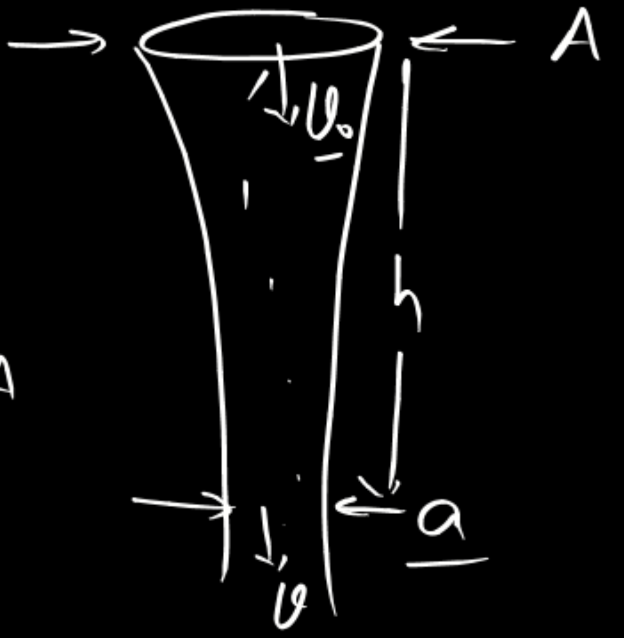
$$\boxed{v' = \frac{2}{7} \omega R}$$

Water is coming out from a tap of cross–section area A with speed v_0. Area of cross–section of water stream changes as it moves down. Which of the following graph can best represent cross–section area 'a' of stream with depth h :

$$a = \frac{A v_0}{\sqrt{v_0^2 + 2gh}}$$

$h = 0 \quad \underline{a = A} \qquad h\uparrow \quad a\downarrow$



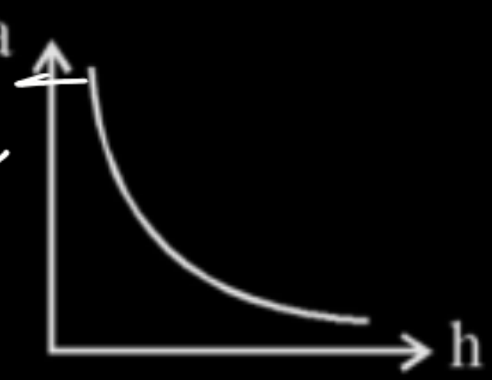


$A v_0 = a v$

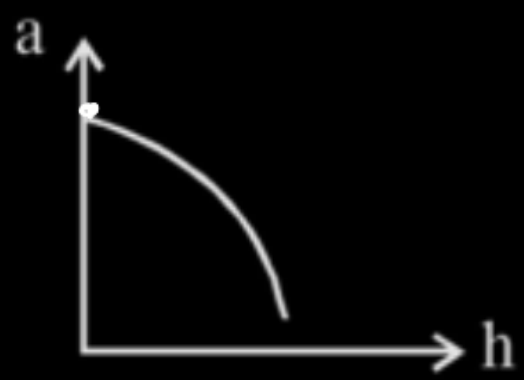
$A v_0 = a\sqrt{v_0^2 + 2gh}$

$P_{atm} + \rho g h + \frac{1}{2}\rho v_0^2 = P_{atm} + 0 + \frac{1}{2}\rho v^2$

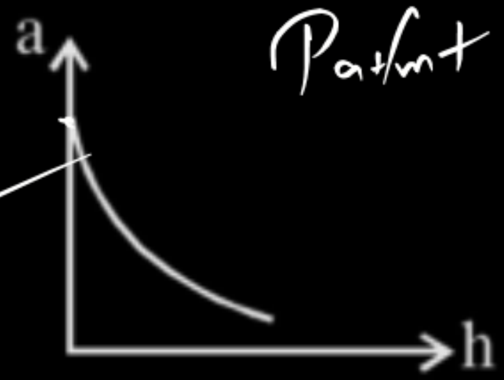
~~(A)~~



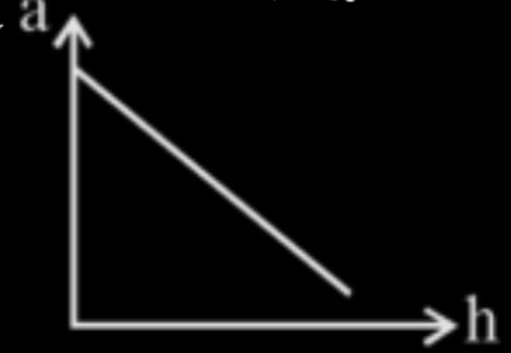
(B)



(C)



~~(D)~~



$$a = AU_0 (U_0^2 + 2gh)^{-1/2}$$

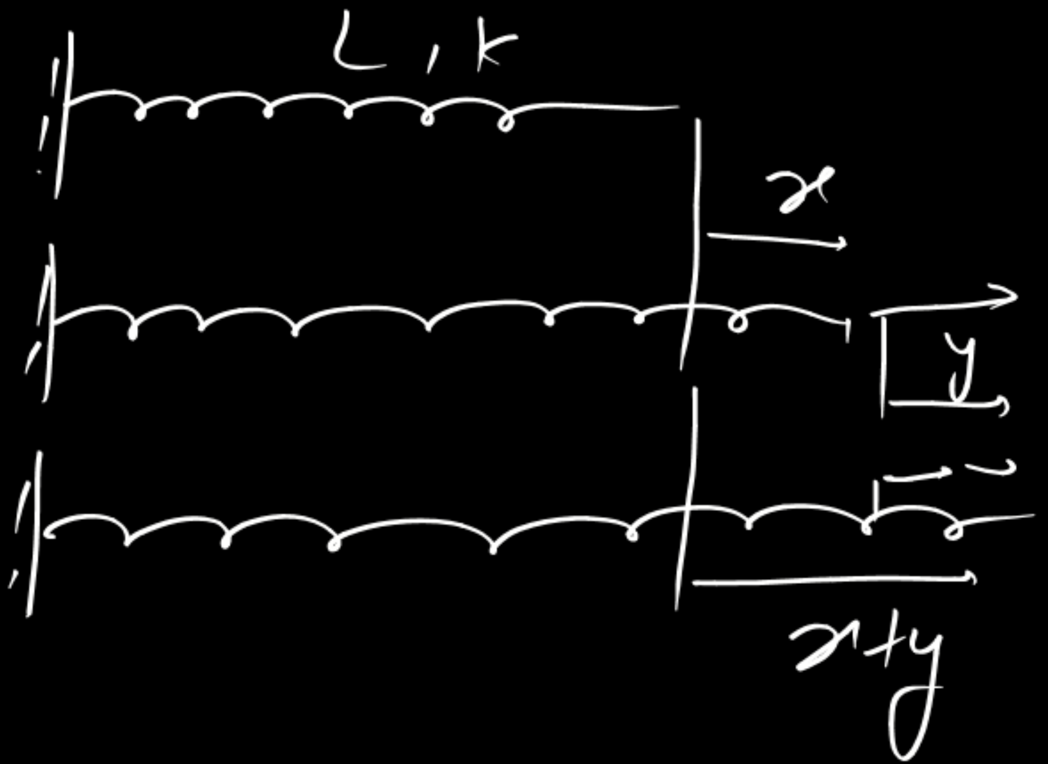
$$\text{slope} = \frac{da}{dh} = AU_0 \frac{-1}{2} (U_0^2 + 2gh)^{-3/2} \times 2g$$

$$\text{slope} = -\frac{AU_0 g}{(U_0^2 + 2gh)^{3/2}} \rightarrow \downarrow$$

$h \uparrow$

An elastic spring of unstretched length ‘L’ and force constant ‘K’ is stretched by a small amount ‘x’. It is further stretched by another small length ‘y’. The work done in second stretching is

(A) $\frac{1}{2}Ky^2$ (B) $\frac{1}{2}K\left(x^2+y^2\right)$ (C) $\frac{1}{2}K(x+y)^2$ (D) $\frac{1}{2}Ky(2x+y)$



$W_{ext} = U_f - U_i$

$= \frac{1}{2}k(x+y)^2 - \frac{1}{2}kx^2$

$= \frac{k}{2}\left(y^2 + 2xy\right)$

$W_s = U_i - U_f$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

A particle oscillates according to the equation x = [1 + $\cos^2$ t + $\cos^4$ t] sin 500t. The lowest frequency present in harmonic components is

(A) $\frac{500}{2\pi}$Hz (B) $\frac{504}{2\pi}$Hz (C) $\frac{496}{2\pi}$Hz (D) $\frac{490}{2\pi}$Hz

$$x = \left(1+\cos^2 t + \cos^4 t\right)\sin 500t$$

$$= \left\{1 + \frac{1+\cos 2t}{2} + \left(\frac{1+\cos 2t}{2}\right)^2\right\}\sin 500t$$

$$= \left\{1 + \frac{1}{2} + \frac{1}{2}\cos 2t + \frac{1}{4} + \frac{\cos^2 2t}{4} + \frac{1}{2}\cos 2t\right\}\sin 500t$$

$$\left(\frac{7}{4} + \cos 2t + \frac{1+\cos 4t}{8}\right)\sin 500t$$

$$\left(\frac{15}{8} + \cos 2t + \frac{\cos 4t}{8}\right)\sin 500t$$

$$\frac{15}{8}\sin 500t + \sin 500t\cos 2t + \frac{1}{8}\sin 500t\cos 4t$$

$$\sin A\cos B = \frac{\sin(A+B)+\sin(A-B)}{2}$$

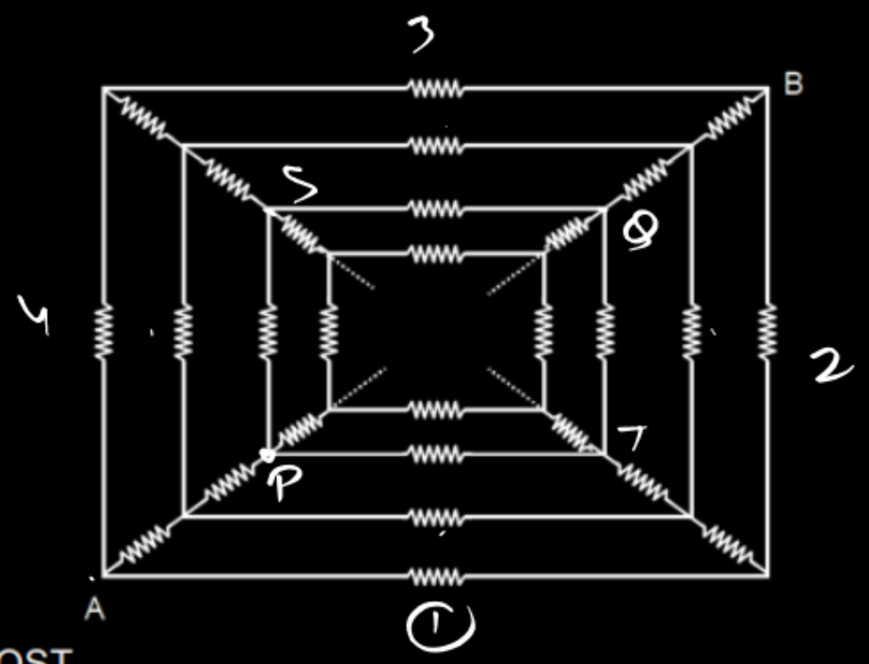
$$\frac{15}{8}\sin 500t + \frac{\sin 502t + \sin 498t}{2} + \frac{1}{8}\,\frac{\sin 504t + \sin 496t}{2}$$

$$\frac{15}{8}\sin 500t + \frac{1}{2}\sin 502t + \frac{1}{2}\sin 498t + \frac{1}{16}\sin 504t + \frac{1}{16}\sin 496t$$

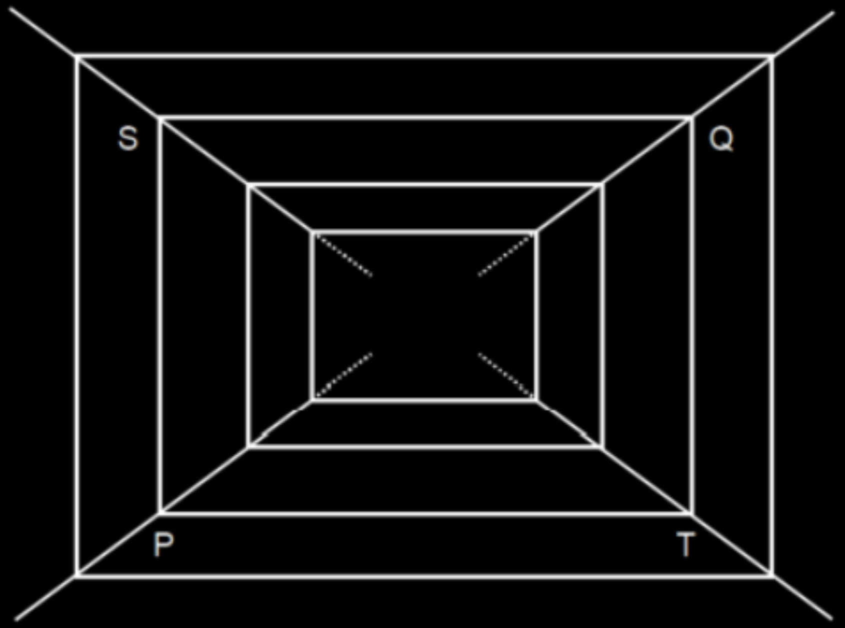
$$496 = 2\pi f$$

$$\boxed{f = \frac{496}{2\pi}}$$

Consider the given circuit made up of infinite resistances, each of resistance R. Each branch represents a resistance.

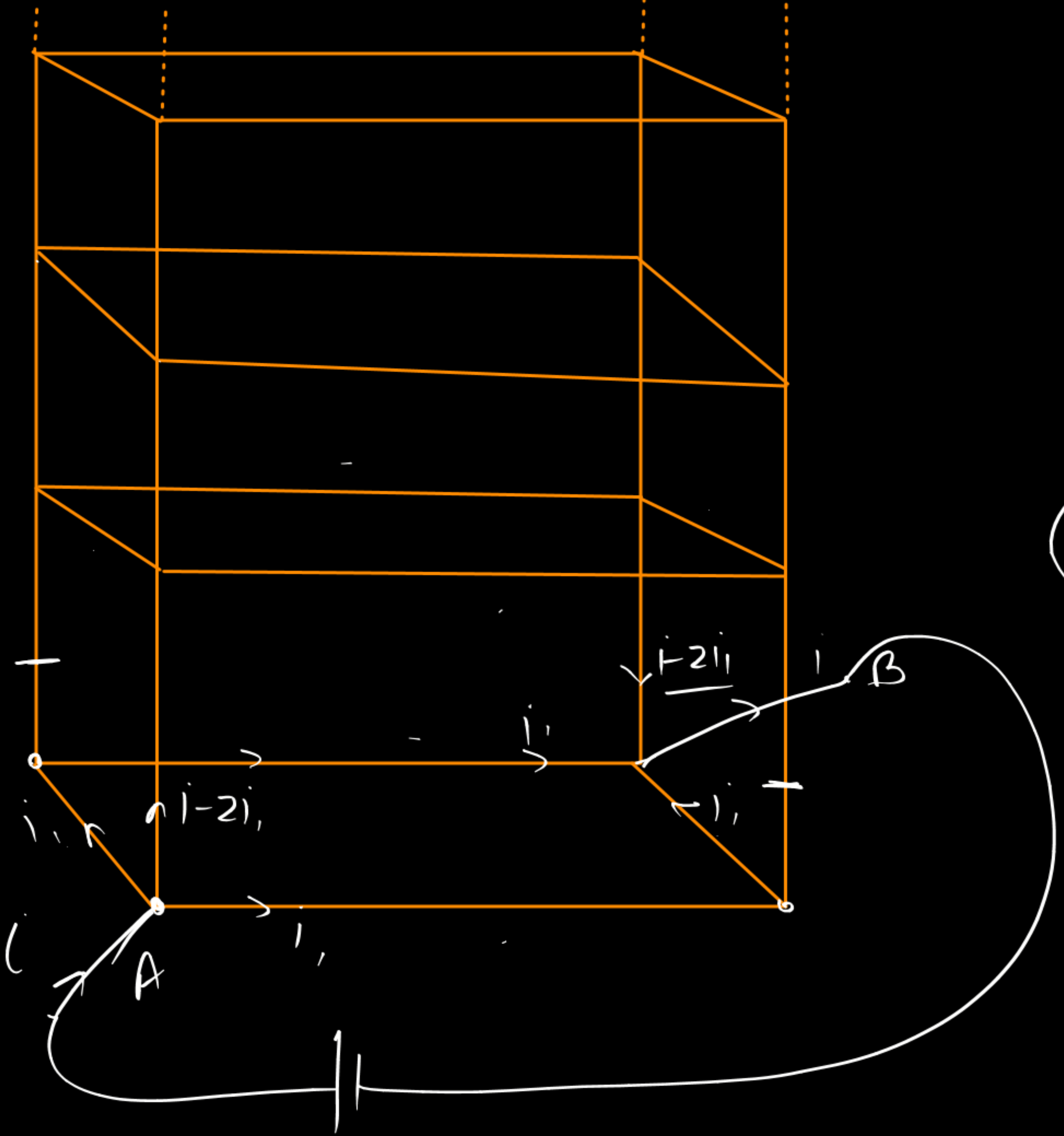


Consider a section PQST



There are infinite resistance outside PQST & infinite resistance inside PQST. Equivalent resistance between PQ is given by

(A) $\frac{R}{\sqrt{2}}$ (B) $\frac{R}{\sqrt{3}}$ (C) $\frac{R}{2}$ (D) $\frac{R}{\sqrt{5}}$

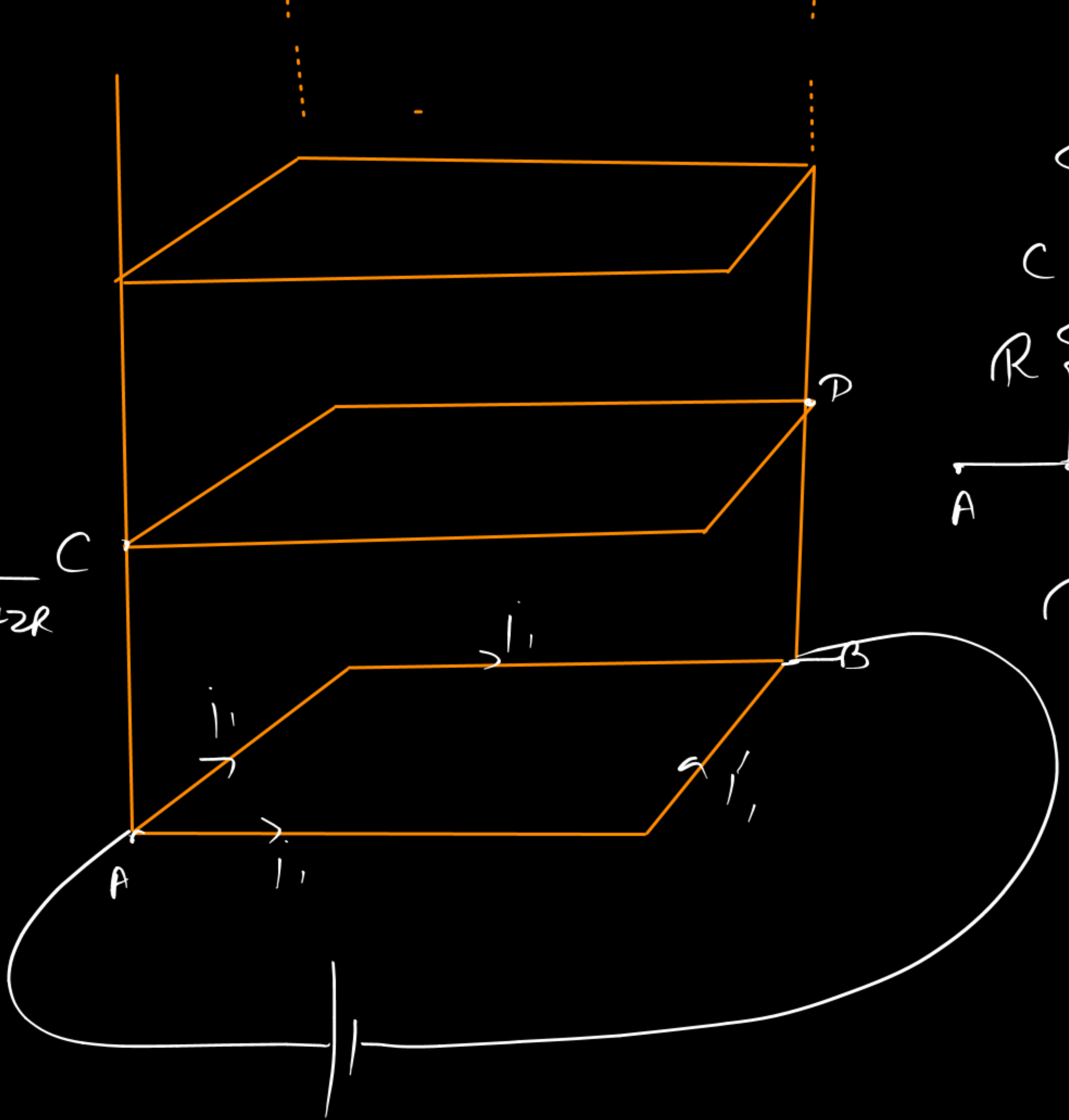


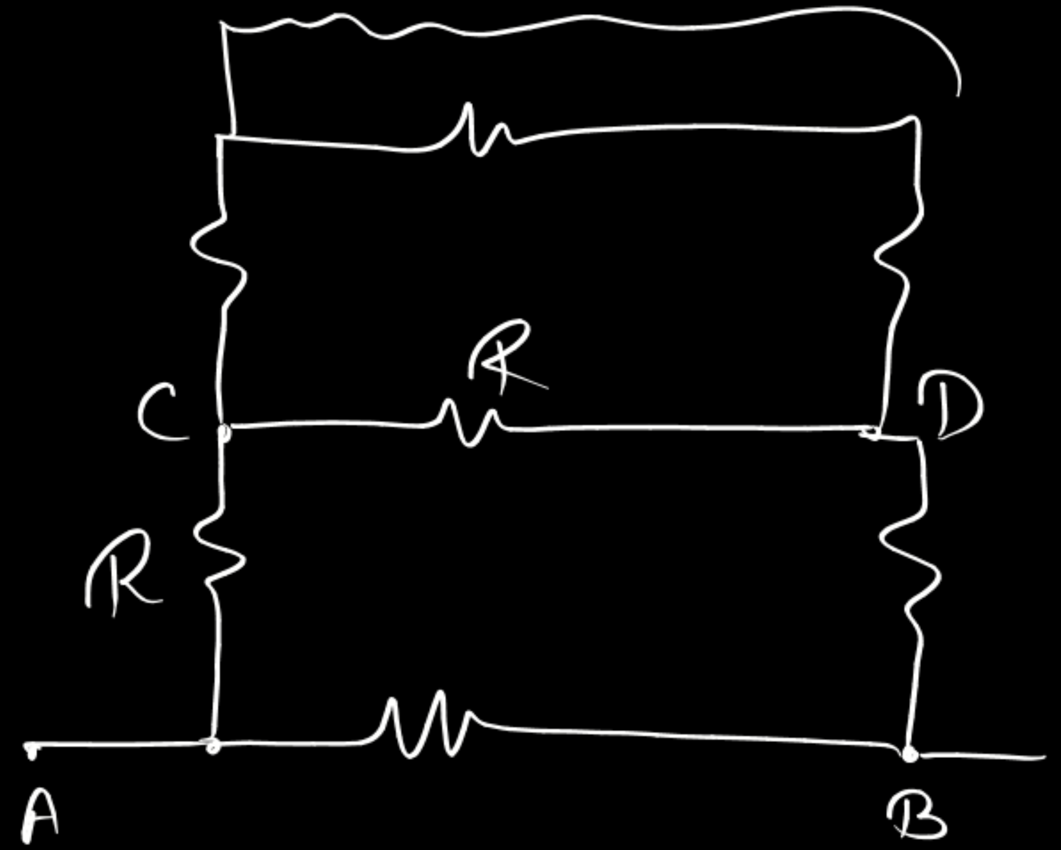
$(\sqrt{3}-1)R$

$R_{eq} + 2R$

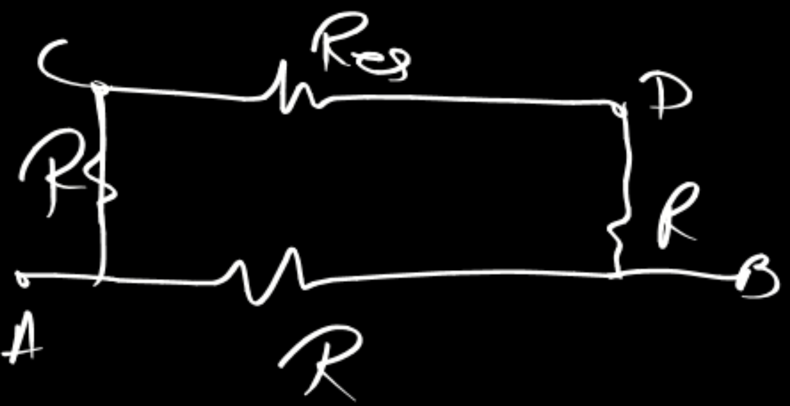
$$\frac{1}{R_{AB}} = \frac{1}{R} + \frac{1}{R_{eq}+2R}$$

$$= \frac{1}{R_{eq}}$$





$$R_{AB} = R_{eq} = R_{CD}$$



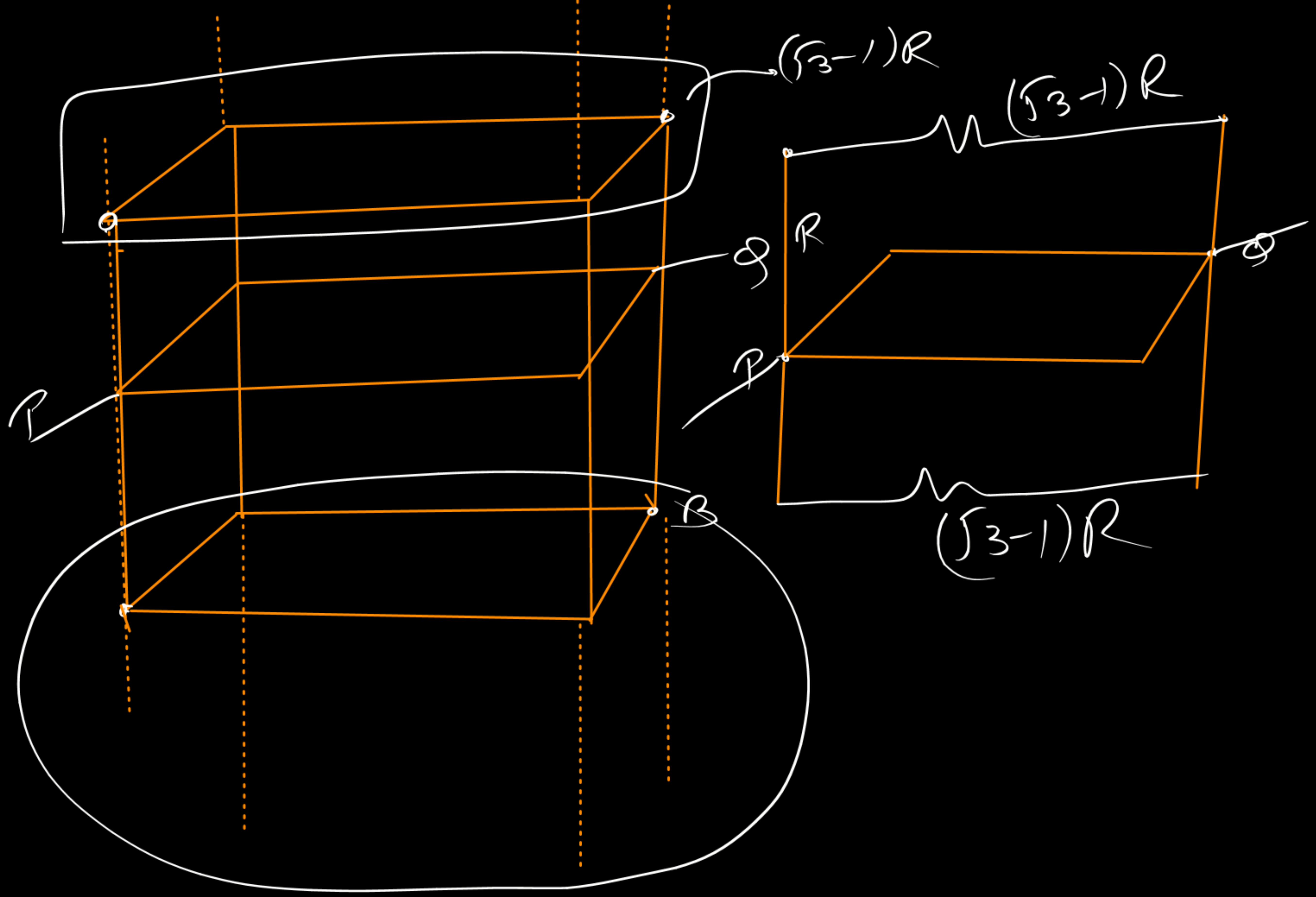
$$\frac{R_{eq}+3R}{(R_{eq}+2R)\,R} = \frac{1}{R_{eq}}$$

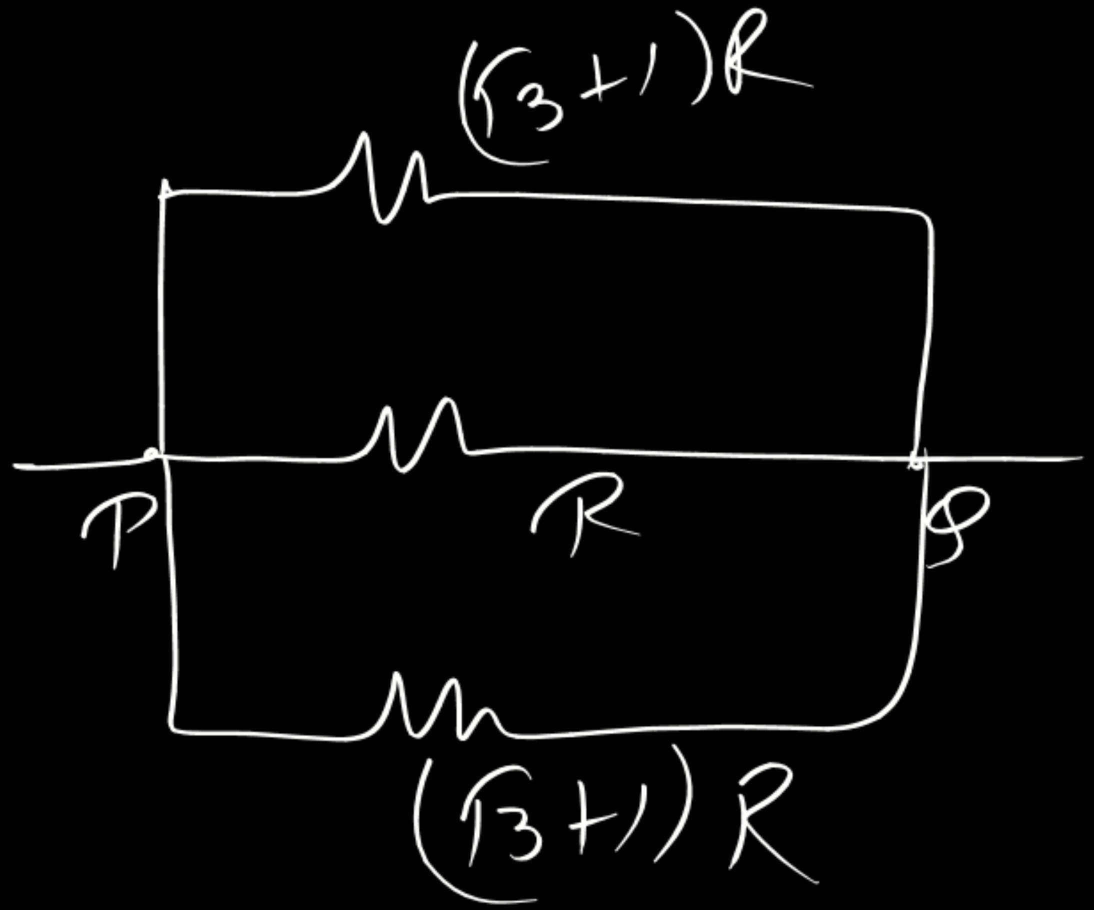
$$R_{eq}^2 + 3R_{eq}R = R_{eq}R + 2R^2$$

$$R_{eq}^2 + 2R_{eq}R - 2R^2 = 0$$

$$R_{eq} = \frac{-2R + \sqrt{4R^2 + 4 \times 1 \times 2R^2}}{2}$$

$$= \underline{-R + \sqrt{3}R} = (\sqrt{3}-1)R$$


(√3−1)R
(√3−1)R
R
Q
Q
P
P
B
(√3−1)R



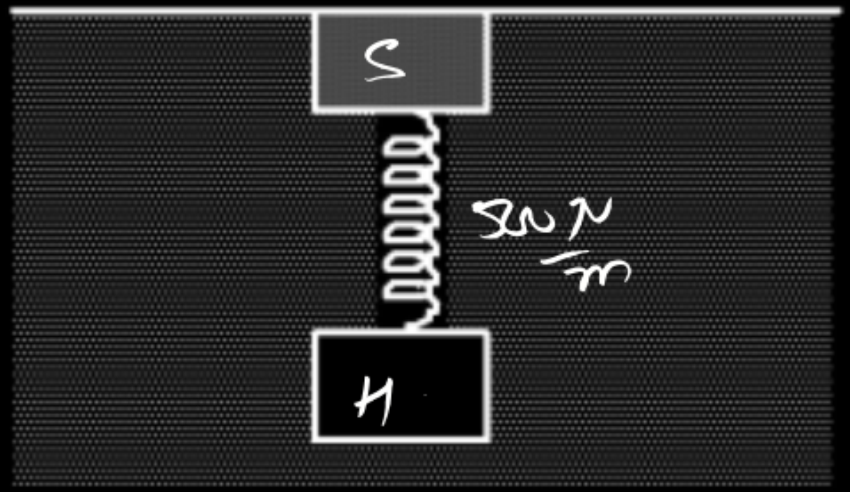
$$\frac{1}{R_{PQ}} = \frac{1}{R} + \frac{1 \times 2}{(\sqrt{3}+1)R}$$

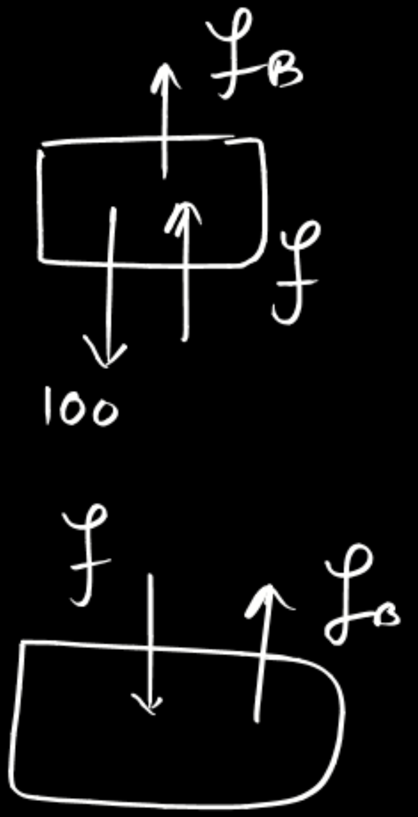
$$= \frac{(\sqrt{3}+1) + 2}{(\sqrt{3}+1)R}$$

$$R_{PQ} = \frac{(\sqrt{3}+1)R}{(\sqrt{3}+3)}$$

$$= \frac{R}{\sqrt{3}} \quad \underline{\text{Ans}}$$

A block of mass 10 kg connected to another hollow block of same size and negligible mass, by a spring of spring constant 500 N/m, floats in water as shown in the figure. The compression in the spring is ($\rho_{water} = 1 \times 10^3 kg/m^3, g = 10 m/s^2$)





~~(A)~~ 10 cm (B) 20 cm (C) 50 cm (D) 100 cm

$$f_B + f = 100$$

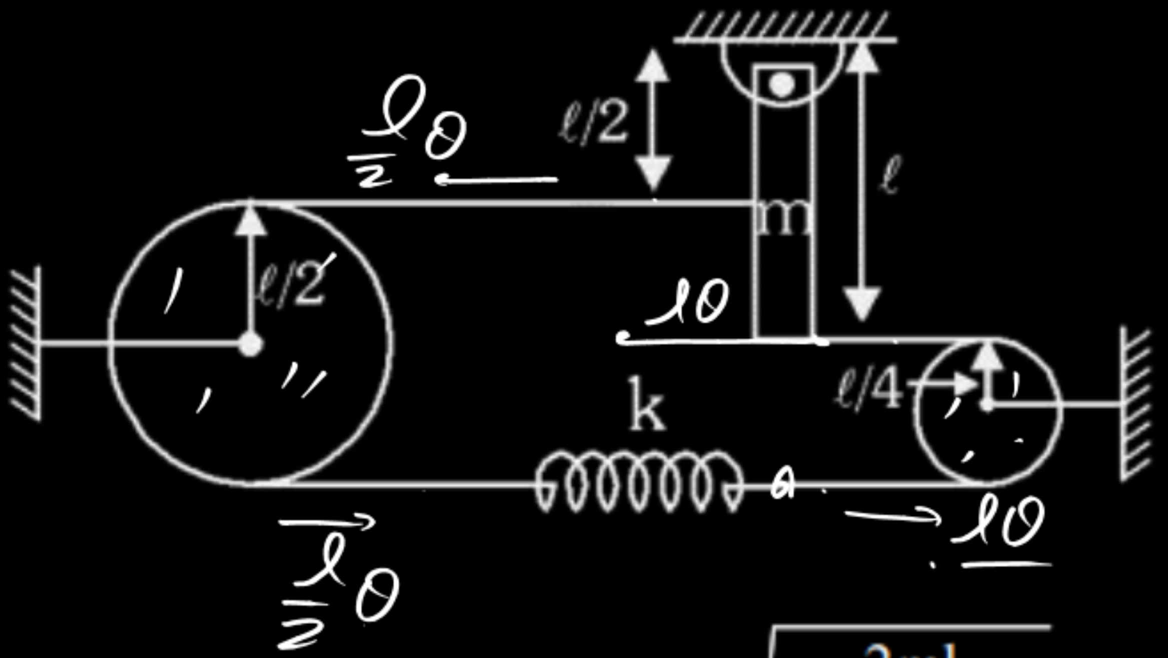
$$\underline{f_B = f} \qquad f_B = f = 50\,\cancel{N} = 500\,\frac{\cancel{N}}{m}\,x$$

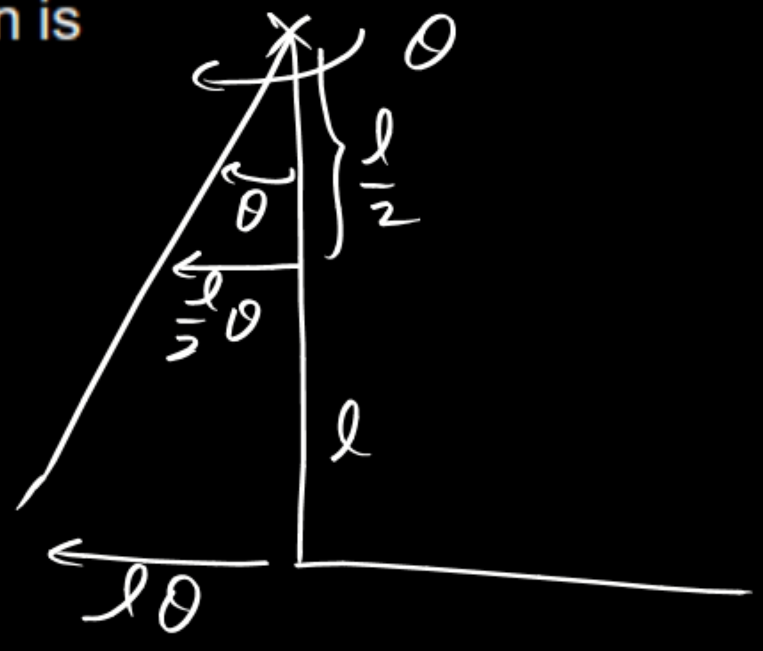
$$\boxed{x = \frac{1}{10}\,m = 10\,cm}$$

A light inextensible string carrying a spring [as shown in figure] is passed over two smooth fixed pulleys. The rod is slightly displaced (about the hinge) from it's equilibrium position (when the spring is unscratched). The radius of smaller pulley is $\frac{\ell}{4}$. The time period for small oscillation is

Vef

$x = \frac{l}{2}\theta$ $\boxed{T = k\frac{l}{2}\theta}$



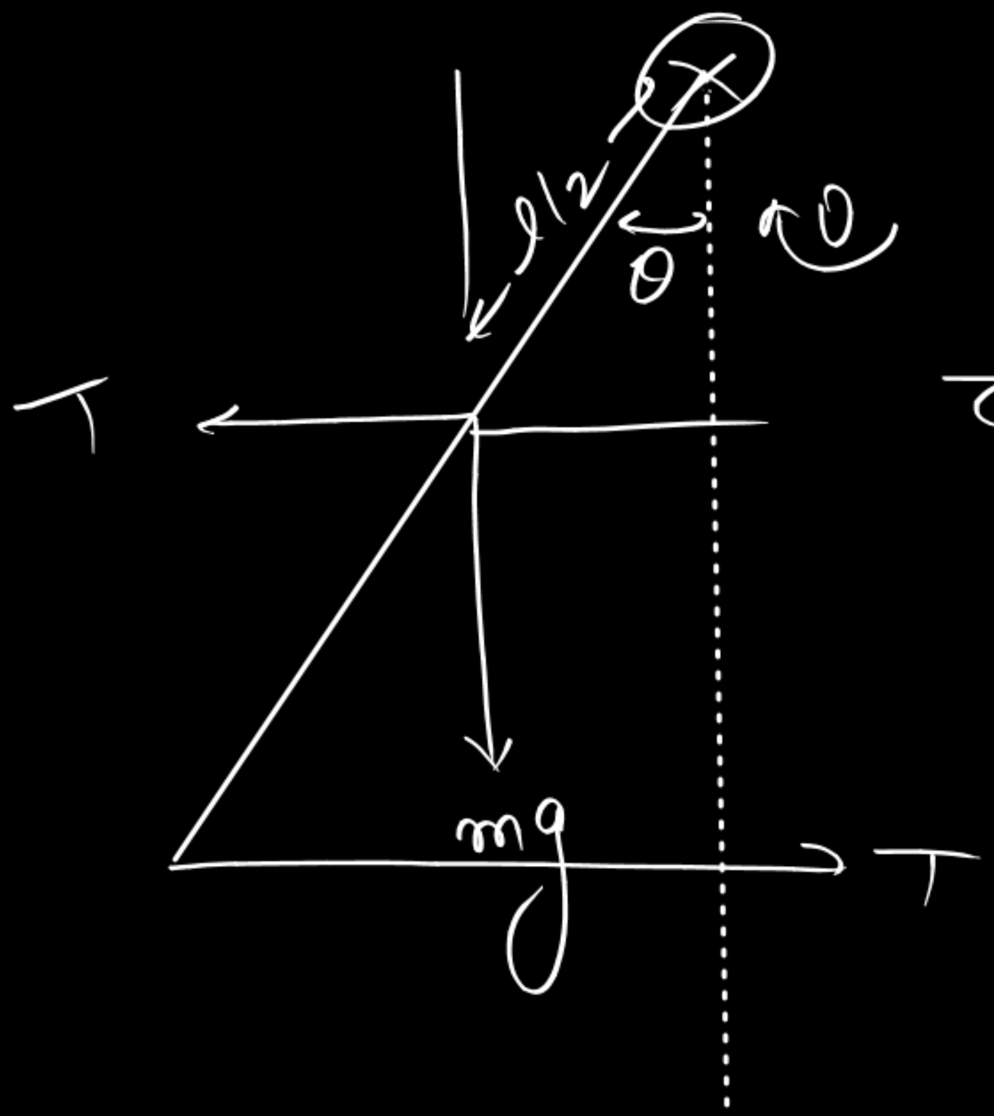


(A) $\pi\sqrt{\frac{ml}{3K\ell + 2mg}} + \pi\sqrt{\frac{2\ell}{3}}$

(B) $\pi\sqrt{\frac{2ml}{3K\ell + 2mg}} + \pi\sqrt{\frac{2\ell}{3}}$

(C) $\pi\sqrt{\frac{ml}{3K\ell + 2mg}} + 2\pi\sqrt{\frac{\ell}{3}}$

(D) $\pi\sqrt{\frac{4ml}{3K\ell + 2mg}} + 2\pi\sqrt{\frac{2\ell}{3}}$



$$\tau = Tl + mg\frac{l}{2}\theta - T\frac{l}{2}$$

$$= T\frac{l}{2} + mg\frac{l}{2}\theta = k\frac{l}{2}\theta\cdot\frac{l}{2} + mg\frac{l}{2}\theta$$

$$\tau = \left(\frac{kl^2}{4} + \frac{mgl}{2}\right)\theta = \frac{ml^2}{3}\alpha$$

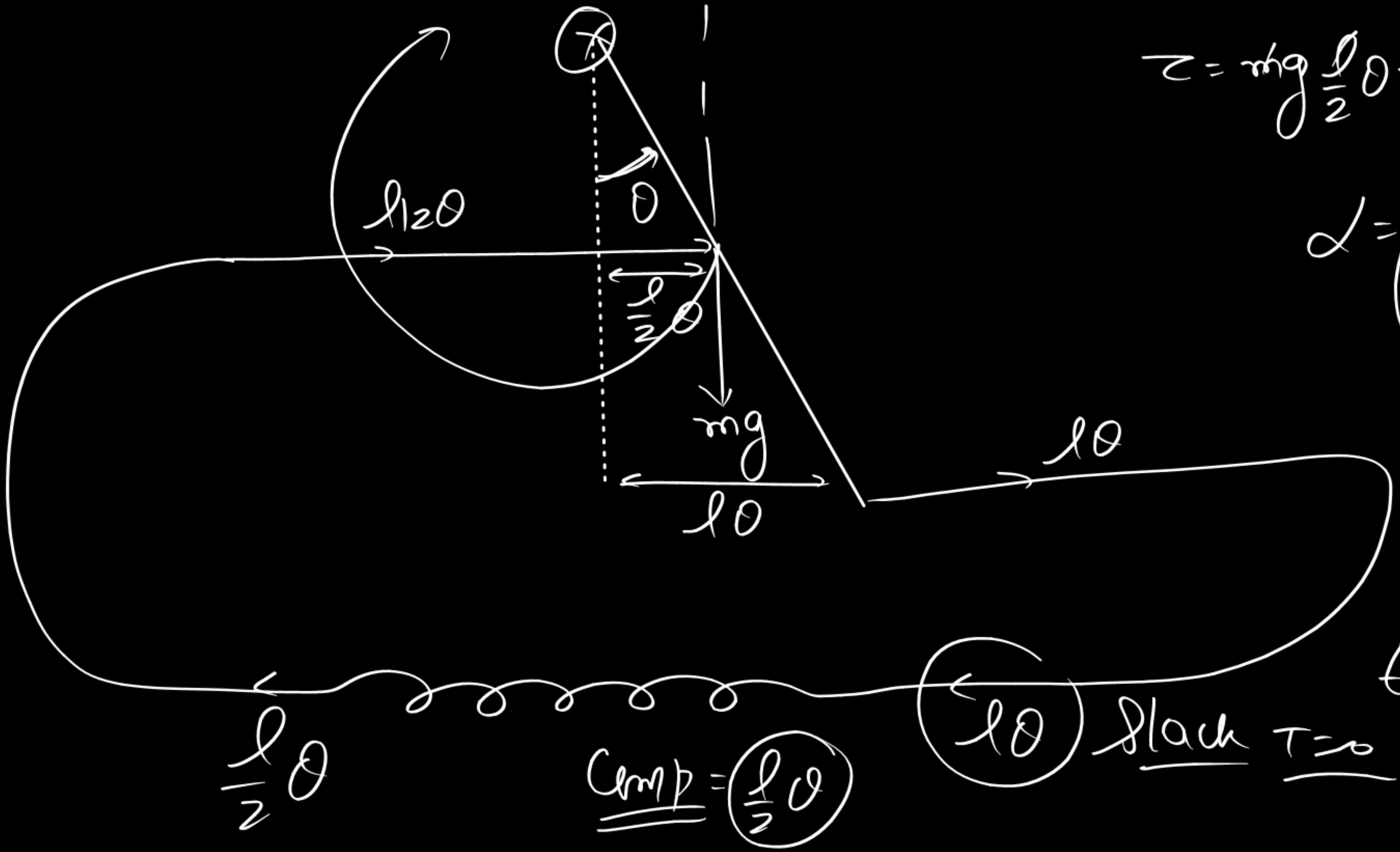
$$\alpha = \underbrace{\left(\frac{3k}{4m} + \frac{3g}{2l}\right)}_{\omega^2}\theta$$

$$\omega = \sqrt{\frac{3kl + 6gm}{4ml}}$$

$$t_1 = \frac{T}{2} = \frac{\pi}{\omega}$$

$$= \pi\sqrt{\frac{4ml}{(3kl+6mg)}}$$

$$\boxed{T = t_1 + t_2}$$



$$\tau = mg\frac{l}{2}\theta = \frac{ml^2}{3}\alpha$$

$$\alpha = \frac{3g}{2l}\theta$$

$$\frac{3g}{2l} \to \omega'^2$$

$$T' = \frac{2\pi}{\omega'}$$

$$t_2 = \frac{\pi}{\omega'} = \pi\sqrt{\frac{2l}{3g}} = t_2$$

$y_1 = 8 \sin(\omega t - kx)$ and $y_2 = 6 \sin(\omega t + kx)$ are two waves travelling in a string of area of cross-section s and density ρ. These two waves are superimposed to produce a standing wave, (partial standing wave) The total amount of energy crossing through a node per second is

(A) $\frac{2\rho\omega^3 s}{3k}$ (B) $\frac{2\rho\omega^3 s}{5k}$ (C) $\frac{2\rho\omega^3 s}{k}$ (D) $\frac{3\rho\omega^3 s}{2k}$

$f = \frac{\omega}{2\pi}$

$$y = y_1 + y_2 = 8\sin(\omega t - kx) + 6\sin(\omega t + kx)$$

$$= 2\sin(\omega t - kx) + 6\{\sin(\omega t - kx) + \sin(\omega t + kx)\}$$

$$y = \underbrace{2\sin(\omega t - kx)}_{T.W.} + \underbrace{12\sin\omega t\cos kx}_{S.S}$$

$P = 0$

$$P = 2\pi^2 f^2 A^2 \mu v$$

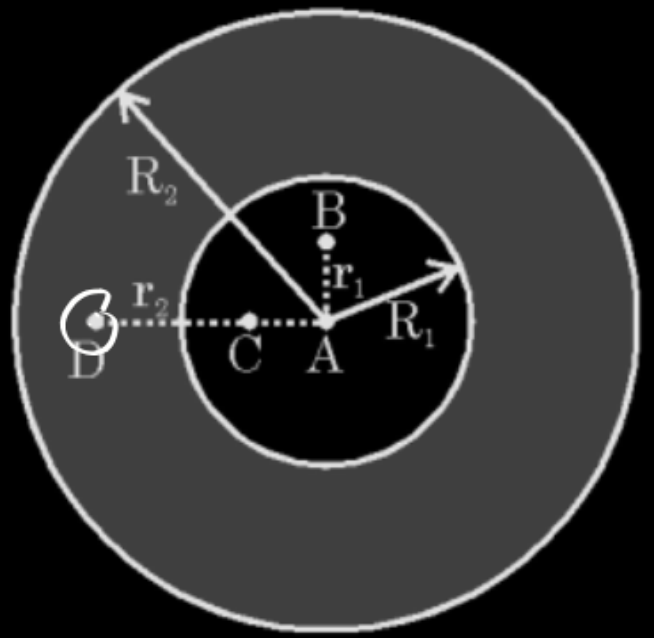
$$= \frac{\omega^2}{2} \times 4 \times \rho s \frac{\omega}{k}$$

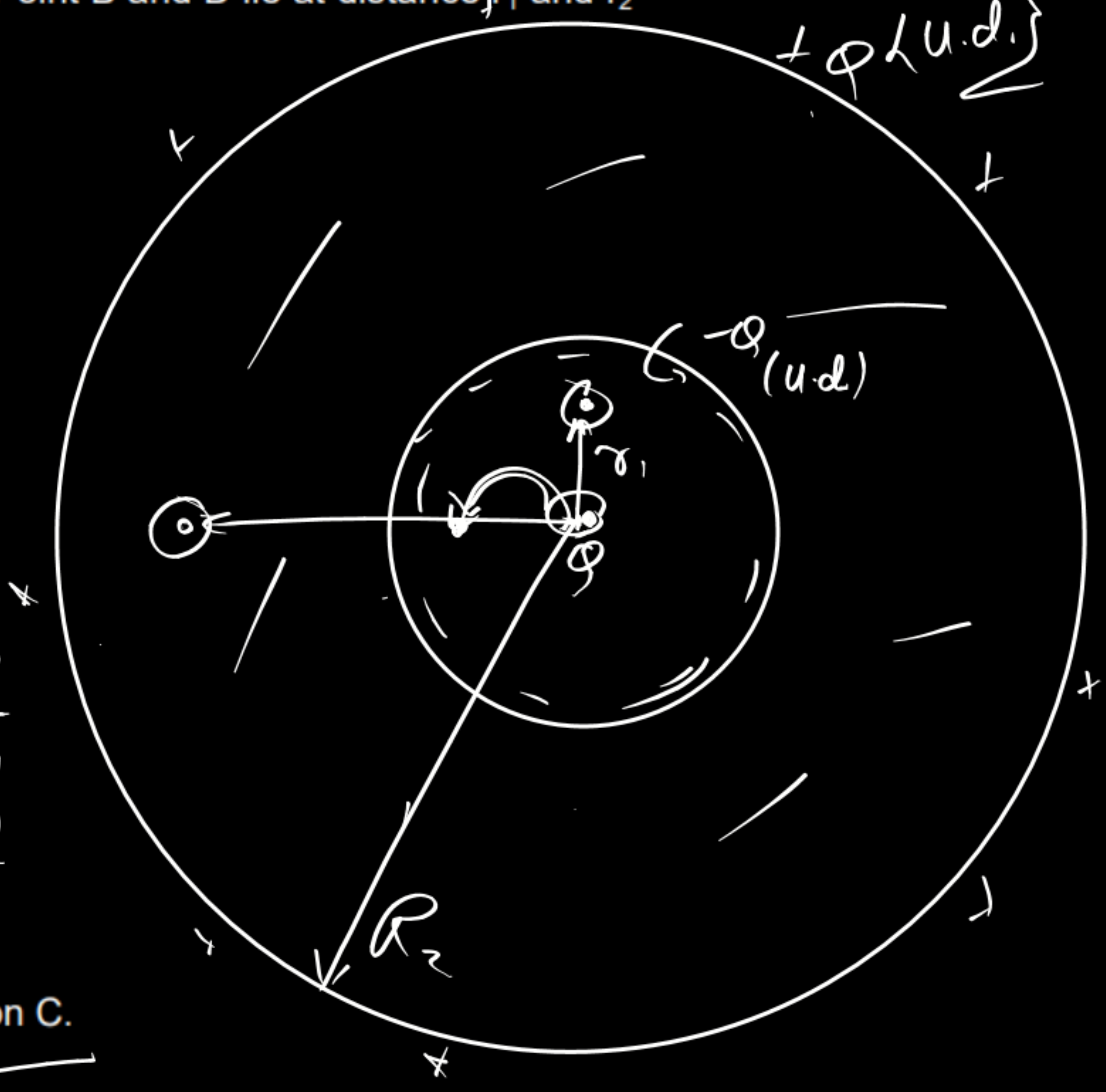
$$= \boxed{\frac{2\omega^3 \rho s}{k}}$$

Consider a thick conducting shell of inner radius R_1 and outer radius R_2 as shown in the figure. Shell is neutral and a point charge Q is kept at point A i.e. centre of shell. Point B and D lie at distance r_1 and r_2 respectively from centre. Choose **CORRECT** option(s) :

$$V_D = \frac{kQ}{R_2} + \cancel{\frac{kQ}{r_2}} + \cancel{\frac{k(-Q)}{r_2}}$$

$$= \frac{kQ}{R_2}$$





(A) Potential at point D is $\frac{kQ}{R_2} + \frac{kQ}{r_2}$

$$V_B = \frac{kQ}{R_2} + \frac{kQ}{r_1} + \frac{k(-Q)}{R_1}$$

(B) Potential at point B is $\frac{kQ}{R_2} + \frac{kQ}{r_1} - \frac{kQ}{R_1}$

(C) Potential at D will change if point charge shifted to position C

(D) Potential at B will not change if point charge is shifted to position C.

A partition divides a container having insulated walls into two compartments I and II. The same gas fills the two compartments whose initial parameters are given. The partition is a conducting wall which can move freely without friction. Which of the following statement is/are correct, with reference to the final equilibrium position ?

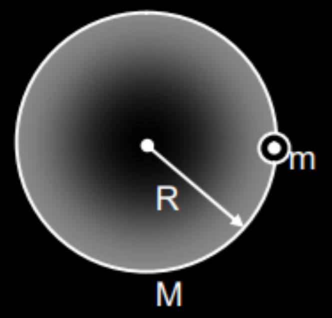
(A) The Pressure in the two compartments are equal

(B) Volume of compartment I is $\frac{3V}{5}$

(C) Volume of compartment II is $\frac{12V}{5}$

(D) Final pressure in compartment I is $\frac{5P}{3}$

A circular ring of mass M and radius R lies on a smooth horizontal surface. An insect of mass m starts moving round the ring with uniform velocity v relative to the ring. Choose the correct option(s): (Use M = m)



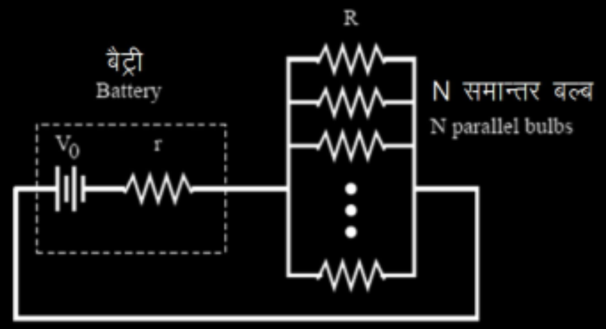
(A) Angular velocity of ring with respect to the ground is $\frac{v}{3R}$

(B) Angular velocity of ring with respect to the ground is $\frac{v}{2R}$

(C) Speed of insect with respect to the ground is $\frac{v}{3}$

(D) Speed of insect with respect to the ground is $\frac{v}{2}$

A certain electric battery is not quite ideal. It can be thought of as a perfect cell with constant output voltage V0 connected in series to a resistance r, but there is no way to remove this internal resistance from the battery.



The battery is connected to N identical light bulbs in parallel. The bulbs each have a fixed resistance R, independent of the current through them. It is observed that the configuration dissipates more total power through the bulbs when N = 5 than it does for any other value of N. Select correct alternative.

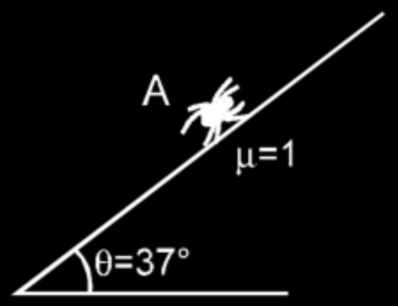
(A) r must be $\frac{R}{5}$

(B) r must be more than $\frac{R}{5}$

(C) r must be less than $\frac{R}{5}$

(D) r may be more than $\frac{R}{5}$

An insect of mass m, starts moving on a rough inclined surface from point A. As the surface is very sticky, the coefficient of friction between the insect and the incline is $\mu = 1$. Assume that it can move in any direction; up the incline or down the incline then ($g = 10\ m/s^2$) : Choose the correct options :



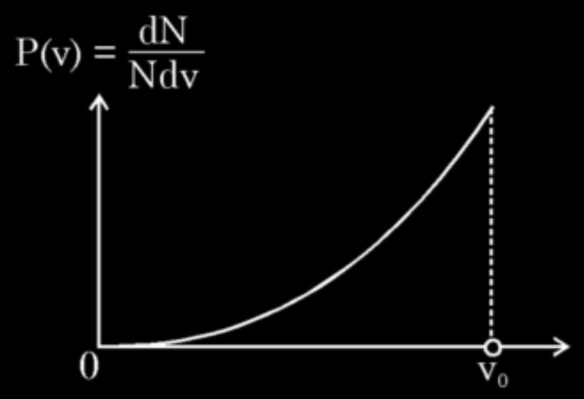
(A) The maximum possible acceleration of the insect can be 14 m/sec^2

(B) The maximum possible acceleration of the insect can be 2 m/sec^2

(C) The insect can move with a constant velocity

(D) no friction force will act on insect when it moves with constant velocity

Figure shows a hypothetical speed distribution for particles of a certain gas : $P(v) = Cv^2$ for $0 < v \leq v_0$ and $P(v) = 0$ for $v > v_0$, where $P(v)$ represent the probability of the particle at a speed of v ($P(v) = \frac{dN}{Ndv}$) and C is some constant.



(A) The value of C is equal to $\frac{1}{v_0^2}$

(B) The value of C is equal to $\frac{3}{v_0^3}$

(C) The average speed of particles is $\frac{3}{4}v_0$

(D) The rms speed of particles is $\sqrt{\frac{3}{4}}v_0$

A hydrogen atom in a state having a binding energy of 0.85 eV, makes a transition to a state with an excitation energy of 10.2 eV. Choose correct options :

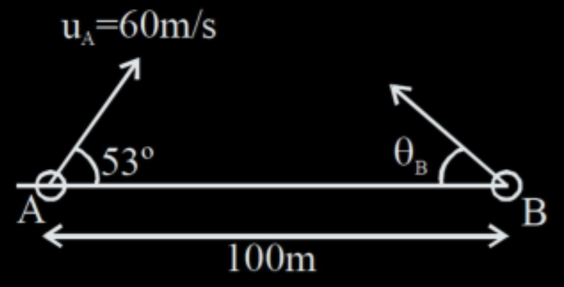
(A) Energy of emitted photon is 2.55 eV.

(B) Jump takes place from n = 4 to n = 2

(C) Energy of emitted photon is 10.2 eV.

(D) Jump takes place from n = 2 to n = 1

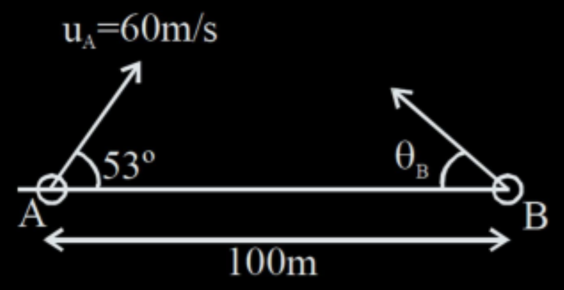
Two particles A and B are projected in same vertical plane as shown in the figure. Their initial position (t = 0), initial speed and angle of projection are indicated. Initially A and B are on a horizontal line



If initial angle of projection $\theta_B = 37°$, what should be initial speed of projection of particle B, so that it hits particle A

(A) 80 m/s (B) 75 m/s (C) 40 m/s (D) 45 m/s.

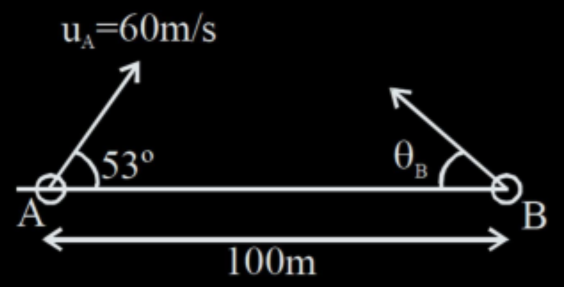
Two particles A and B are projected in same vertical plane as shown in the figure. Their initial position (t = 0), initial speed and angle of projection are indicated. Initially A and B are on a horizontal line



An observer on particles A observes that particle B is moving at angle 8° with horizontal in downward direction. Initial angles of projection of each particle remain same as previous question. Speed of particle B is :

(A) 80 m/s (B) 75 m/s (C) 60 m/s (D) 40 m/s.

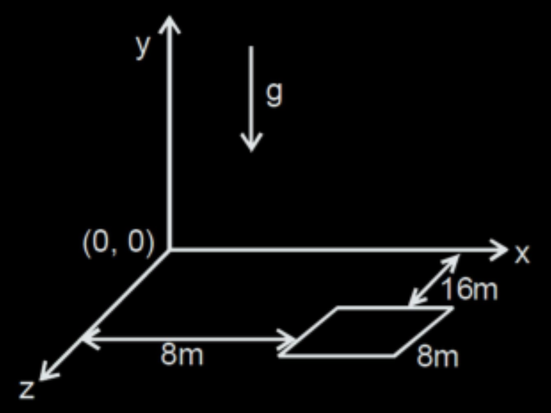
Two particles A and B are projected in same vertical plane as shown in the figure. Their initial position (t = 0), initial speed and angle of projection are indicated. Initially A and B are on a horizontal line



In the situation of previous question, when observer on particle A, observes that particle B appears to be moving at 8° below horizontal (θ_B = 37°). After how long (in seconds) distance between A and B is minimum. (t = 0 situation is shown in figure)

(A) $\frac{3\sqrt{2}}{5}\sin 8^\circ$ (B) $\frac{3\sqrt{2}}{5}\cos 8^\circ$ (C) $\frac{5}{3\sqrt{2}}\sin 8^\circ$ (D) $\frac{5}{3\sqrt{2}}\cos 8^\circ$

A square platform of side length 8 m is situated in x–z plane such that it is at 16 m from the x–axis and 8 m from the z-axis as shown in figure. A particle is projected with velocity $\vec{V} = (v_2\hat{i} + 25\hat{j})\,m/s$ relative to wind from origin and at the same instant the platform starts with acceleration $\vec{a} = (2\hat{i} + 2.5\hat{j})\,m/s^2$. Wind is blowing with velocity $v_1\hat{k}$. ($g = 10\ m/s^2$)

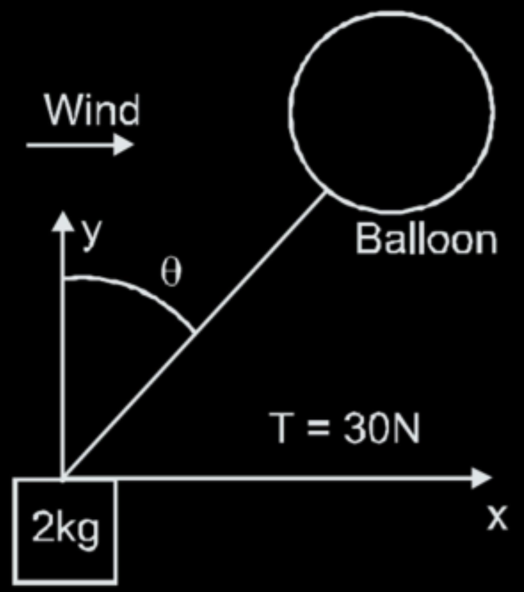


List-I	List-II
(P) Least possible values of v_2 (in m/s) so that particle hits the platform or edge of platform is	(1) 4
(Q) Least possible value of v_1 (in m/s) so that particle hits the platform or edge of platform is	(2) 6
(R) If t is the time (in second) after particle hits the platform then 2t is equal to	(3) 8
(S) Value of displacement with respect to ground (in m) of the particle in y–direction, when v_2 has its minimum possible value is (till particle hits the platform or edge of platform)	(4) 20

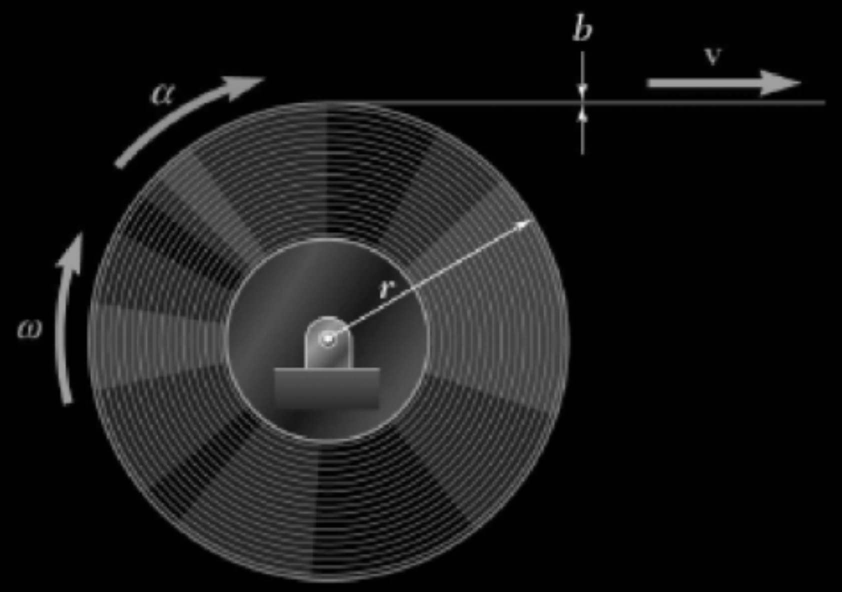
Codes :

	P	Q	R	S
(A)	2	4	3	1
(B)	2	1	3	4
(C)	2	3	4	1
(D)	2	1	4	3

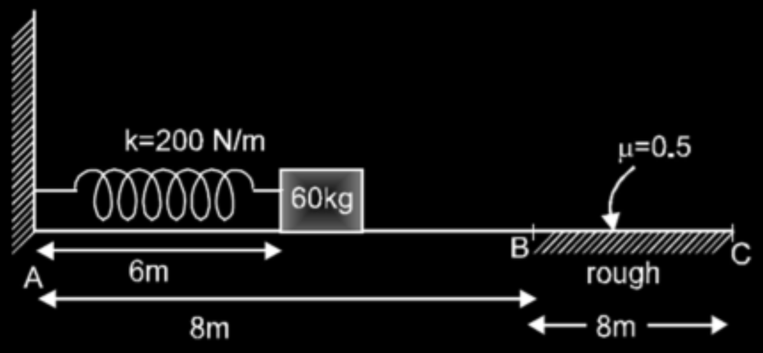
A balloon is tied to a block. The mass of the block is 2kg. The tension of the string between the balloon and the block is 30N. Due to the wind, the string has an angle θ relative to the vertical direction. $\cos\theta = 4/5$ and $\sin\theta = 3/5$. Assume the acceleration due to gravity is $g = 10\ m/s^2$. Also assume the block is small so the force on the block from the wind can be ignored. If the x-component and the y-component of the acceleration a of the block are 'm' and 'n' (in m/s^2) respectively, then the value of $|m-n|$ is :



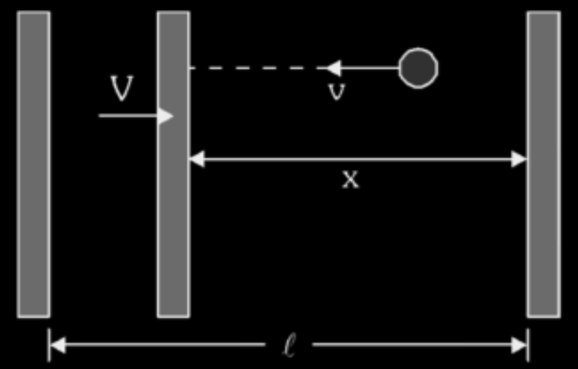
In a continuous printing process, paper is drawn into the presses at a constant speed v Denoting by r the radius of the paper roll at any given time and by b the thickness of the paper, angular acceleration of the paper roll is $\frac{bv^2}{n\,\pi r^3}$. Find the value of n.



A block of mass 60 kg attached to the spring as shown in figure is released from rest when compression in the spring is 2m (natural length of spring is 8m). Surface AB is smooth while surface BC is rough. Block travels x distance before coming to complete rest. Value of x is : [g = 10 m/s^2]



A "super ball" of mass m bounces back and forth with speed v be tween two parallel walls, as shown. The walls are initially separated by distance ℓ. Gravity is neglected and the collisions are perfectly elastic.



If one surface is slowly moved toward the other with speed V << v, the bounce rate will increase due to the shorter distance between collisions, and because the ball's speed increases when it bounces from the moving surface. If the average force at distance x is $\frac{mv_0^2\ell^2}{nx^3}$. Find the value of n.

In the situation shown coefficient of friction between A and B is 0.5 and between B and C is 0.3. Friction acting between B and C is $\frac{7x}{9}$ Newton then x is :